



**CASE WESTERN RESERVE
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May 22, 2025

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Dear Professor Landry,

Thank you for taking the time to edit our manuscript “Comprehensive mutational characterization of the calcium-sensing STIM1 EF-hand reveals residues essential for structure and function” (GENETICS -2025 -307871). We are appreciative of the rigorous and thoughtful reviews and their comments for improvement. We are also heartened by the positive assessments provided by the reviewers, with Reviewer 1 noting that they enjoyed reading the manuscript, Reviewer 2 remarking that the manuscript is well written and the results “clear, not over-interpreted, and reflect a careful underlying experimental design”, and Reviewer 3 noting that it was a “well-written and significant contribution to the field of functional genomics”, “it offers valuable mechanistic insights”, and “The dataset generated provides a valuable resource for future studies”. We are appreciative of the reviewers understanding that STIM1 is a challenging, albeit worthwhile, protein to study. The reviews were thorough, and combined for a total of 30 distinct points for improvement, which we systematically addressed as detailed in our point-by-point response. To summarize the major changes:

- Quantification of the difference in doubling time of STIM1 knockout LP cells as compared to the standard LP cells. Also, quantification of the growth kinetics of the three strains of E.coli used in the work.
- Changes to main figure 9.
- Addition of 22 additional supplementary figure panels
- Update of Supplementary table 1 with individual replicate score values, and a new Supplementary table 6 with FoldX, AlphaMissense, and Rosace scores.

We feel we have addressed the concerns raised by the reviewers, and hope that our manuscript is now suitable for publication. Below, we have included a point-by-point discussion of the reviewers’ criticisms. We have also noted the changes made within the manuscript by highlighting the altered text.

Thank you for your consideration,
Nisha D. Kamath and Kenneth A. Matreyek